



Calcium carbonate unit realignment under acidification: A potential compensatory mechanism in an edible estuarine oyster

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ABSTRACT

Ocean acidification (OA) is well-known for impairing marine calcification; however, the end response of several essential species to this perturbation remains unknown. Decreased pH and saturation levels (Ω) of minerals under OA is projected to alter shell crystallography and thus to reduce shell mechanical properties. This study examined this hypothesis using a commercially important estuarine oyster *Magallana hongkongensis*. Although shell damage occurred on the outmost prismatic layer and the undying myostracum at decreased pH 7.6 and 7.3, the major foliated layer was relatively unharmed. Oysters maintained their shell hardness and stiffness through altered crystal unit orientation under pH 7.6 conditions. However, under the undersaturated conditions ($\Omega_{\text{cal}} \sim 0.8$) at pH 7.3, the realigned crystal units in foliated layer ultimately resulted in less stiff shells which indicated although estuarine oysters are mechanically resistant to unfavorable calcification conditions, extremely low pH condition is still a threat to this essential species.

1. Introduction

Edible oyster species in Chinese coastal areas are belonging to the family *Magallana*, previously *Crassostrea*, and are well known for their ecosystem services such as calcareous reef formation on coastal habitats and for seafood production through aquaculture (Lenihan and Peterson, 1998; Wang et al., 2008; Wang et al., 2010; Zhang et al., 2012). The calcareous shells they produce not only protect their soft tissues but also provide hard substratum for many benthic species. Furthermore, the biologically controlled biomineralization process to produce oyster shells has inspired dentists, material scientists and orthopedics (Fujita et al., 1990; Lee et al., 2008a; Shen et al., 2014). Oyster shells are composites of highly oriented crystal units (> 95%) and organic matrix proteins (< 5%) (Lee et al., 2008c). Importantly, the assemblage of these two components within a shell is species-specific (Lee et al., 2008b; Lombardi et al., 2013). Depending on the environment and genotype, species assemble crystal units within organic matrix proteins in different ways to maximize mechanical properties with minimal energy expenditure and weight gain (Menig et al., 2000, 2001; Lin et al., 2006). Oysters have a complex life cycle in which the planktonic veliger larvae develop for about 2 weeks before becoming

pediveliger larvae and metamorphosing into benthic juveniles. The newly attached juveniles form hard and stiff calcareous shells for protection via an energetically expensive and sophisticated biomineralization process, however due to the fragility of these early shells the juvenile oysters remain highly susceptible to predation (Newell et al., 2007). The juvenile oyster's shell is mainly composed of calcite, a less soluble form of calcium carbonate (CaCO_3), and a small amount of aragonite, a more soluble form of CaCO_3 (Weiner and Addadi, 1997). The complex features and diversity of shell crystal assemblages are a rich source of information for the development of new biomaterials. Nevertheless, the focus of this study is on the environmental implications of shell assemblages.

Like many marine calcifying organisms, oysters depend on their environment for biomineralization, and in current and near-future coastal oceans this can prove challenging. One important component of their environment is the seawater carbonate chemistry, and decreasing pH and carbonate ions due to elevated anthropogenic carbon-dioxide is of greatest concern (Caldeira and Wickett, 2003). Understanding the influence of this change on the end-products of the biomineralization (i.e. shells structure and mechanics) is an important scientific knowledge gap. The elevation of CO_2 in coastal waters triggers a series of

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chemical changes through the processes of ocean acidification (OA), which not only depletes resources (i.e. carbonate ion concentration) needed for shell formation but also facilitates the dissolution of formed shells as saturation levels of CaCO_3 minerals in the environment decrease (Bednarsek et al., 2012). Consequently, the shell formation processes of several marine calcifying marine organisms, including oysters, are known to be impeded and impaired by OA (Ries, 2011). According to a recent report, exposure of the Pacific oyster, *Magallana gigas* (previously *Crassostrea gigas*) to decreased-pH condition (pH 7.6) altered their energy allocation strategy, shell structural integrity, and finally, reduced shell hardness (Timmins-Schiffman et al., 2014). However, a similar decreased-pH level did not affect shell hardness in juveniles of the estuarine Eastern oyster species, *Crassostrea virginica* (Dickinson et al., 2012; Ivanina et al., 2013). Despite the economic importance of these oysters and the expected threat to the formation of their shells, researchers are only just beginning to understand the effects of OA on oyster shell structure and mechanics.

An oyster shell comprises four mineralized layers: prismatic, foliated, chalky and myostracum (MacDonald et al., 2010). Within the “prominent” foliated layer which is consist of calcite crystal units in elongated foliated laths, the crystal unit's size, shape, and orientation each play an important role in determining the shell's mechanical strength, and thus, the survival of oyster (Lee et al., 2008b; Lee et al., 2008c; Meng et al., 2018). For example, shells with a microstructure that is formed by interlocking and complex crystal units are much stronger than shells of similar size that are formed with straight crystal units. The strong correlation between crystal unit morphology (and orientation) and mechanical strength is expected because resistance to fracture generation increases with decreasing crystal unit size and orientation complexity, as less external force is required for a crack to propagate through larger crystal units. Therefore, it is important to determine the influence of OA on crystal unit morphology, size and orientation and the resulting mechanical properties.

To cope with decreased environmental pH and carbonate ion concentration or saturation state under OA, oysters might allocate more energy toward maintaining the appropriate pH level at the calcification site and to pumping in adequate carbonate ions, against an increasingly unfavorable concentration gradient (Kurihara, 2008; Hofmann and Todgham, 2010). This shift in energy allocation to counter the physiological imbalance at the calcification site may reduce energy available for production of proper biomineralization tools for the assemblage of crystal units in the shell, which could severely affect the mechanical properties of the shell. For instance, oysters growing in a decreased pH environment may not capable of assembling the mineralized foliated layer properly, which could ultimately reduce the shell hardness and stiffness. However, oysters that inhabit estuarine habitats have higher tolerance ranges in regard to a variety of environmental variables, including pH. Estuarine oysters may, therefore, have physiological or biomineral plasticity allowing them to tolerate decreases in pH, effectively preparing them for the projected environmental change due to OA. In order to explore the tolerance of an estuarine oyster to the CO_2 -driven seawater acidification from a biomineralization perspective, we have investigated the effect of OA by using three environmentally and climatically relevant levels of decreased pH as proxy on the structural features of juvenile oyster shells including the microstructure and crystallography, and the mechanical properties of the commercially important oyster, *M. hongkongensis*.

2. Material and methods

2.1. Experimental animal and design

During the peak reproductive season (June 2016) mature adult oysters (*Magallana hongkongensis*) were collected from an oyster hatchery at the Zhanjiang Research Station (21°20'N, 110°40'E), South China Sea Institute, Chinese Academy of Sciences, China. The brood

stock was acclimatized in the laboratory under ambient water conditions (15 psu salinity, 27 °C and $\text{pH}_{\text{(NBS)}}$ 8.1) for 3 to 4 weeks in large flow-through tanks. Gonads from > 10 males and females were obtained by “strip spawning” (Dineshram et al., 2013). Fertilization and embryonic development occurred in ambient conditions. At 2 weeks post-fertilization, pediveliger larvae were collected and used in the following pH perturbation experiment.

Three environmentally and climatically representative pH levels (pH 8.1, 7.6 and 7.3) were used to determine the effect of decreased pH or high- CO_2 induced ocean acidification (OA) on the microstructure, crystallography and mechanical properties of juvenile shells. Each of the three pH treatments had three replicate cultures. According to IPCC projections, the average pH in coastal areas may drop to pH 7.9 and 7.6 by the years 2100 and 2300, respectively (IPCC, 2014). The selected decreased pH levels are already ecologically relevant, because the pH of the estuarine habitat where *M. hongkongensis* live naturally fluctuates by as much as 0.5 pH units from the ambient level due to river run off and algal respiration (Dineshram et al., 2013; Duarte et al., 2013; Environmental Protection Department, 2016). The pH perturbation procedure and larval culture techniques have been previously described (Dineshram et al., 2013). Briefly, decreased pH conditions were obtained through bubbling CO_2 -enriched air with the appropriate CO_2 concentration. The CO_2 concentration in the bubbled air was adjusted using a dual variable area flow controller (Cole-Parmer Inc.). The pediveliger larvae were randomly distributed among the 12 treatment 5 l culture tanks (control and 3 treatment pH levels $\times$ 3 replicates each, 2–3 larvae ml^{-1} in 5 l buckets, 1 μm filtered seawater, 15 psu salinity, at 27 °C). The tanks were provided with a plastic substrate coated with 7-day-old natural biofilm for attachment and metamorphosis of the pediveliger larvae. The metamorphosed juveniles were transferred 48 h-post-settlement and reared under the same pH level for 12 weeks after metamorphosis. Pediveliger larvae and juveniles were fed ad libitum with mixed algal cultures ($5\text{--}10 \times 10^6$ cells ml^{-1}) of *Isochrysis galbana* and *Chaetoceros gracilis*.

Seawater pH (NBS scale) and the temperature were monitored using a Mettler-Toledo (SG2) probe and salinity with a refractometer (ATAGO, S/Mill0E; Japan). The probe calibration used NIST buffers (pH = 4.01, 7.00, and 9.21; Mettler Toledo, GmbH Analytical CH8603 Schwerzenbach, Switzerland). In each replicate tank, pH, temperature and salinity were measured daily and averaged within and among days. Afterward, by averaging the replicate culture tanks for each treatment ($n = 3$), the treatment level (Mean $\pm$ SD; Table S1) was calculated. Samples of seawater (50 ml) from each culture tank were collected every two weeks and treated with 10 μl of 250 mM mercuric chloride for total alkalinity (TA) analysis using the Alkalinity Titrator (AC-A2, Apollo SciTech's Inc., U.S.). The TA measurement was standardized with a seawater reference material (Batch 106, A.G. Dickson, Scripps Institution of Oceanography, U.S.). The carbonate system parameters, i.e. carbon dioxide partial pressure (pCO_2 ; μatm), carbonate ion concentration (CO_3^{2-} ; $\mu\text{mol kg}^{-1}$), calcite and aragonite saturation states (Ω_{Ca} , Ω_{Ar}), were computed using the CO2SYS software program (Pierrot et al., 2006) with equilibrium constants K_1 , K_2 and KSO_4 (Mehrbach et al., 1973; Dickson and Millero, 1987) (Table S1). At the end of the experimental period, the juveniles were preserved in 75% ethanol for shell analysis.

2.2. Shell preparation

Between the two shell valves of the juvenile oyster, the right (or top) valve is responsible for the protection of the soft body from predators and the environmental. Therefore, the top valve was used in this study to exam the microstructure, crystallography and mechanical properties of the oyster shell using Scanning Electron Microscope imaging (SEM), Electron Back Scatter Diffraction (EBSD) and nanoindentation, respectively. Oyster shells were obtained by carefully removing the soft tissues without damaging the internal surface and airdried before

embedding by epoxy resin (EpoxyCure, Buehler). The oyster shell is composed of four layers: prismatic, folia, chalky and myostracum. Among these layers, the foliated layer is a major bulk portion of the shell (Lee et al., 2008b). Therefore, we targeted the foliated layer in examining the effect of decreased pH. The shell was embedded in resin blocks which were then sectioned longitudinally and polished with grit papers (P320, P800, P1200, P2500 and P4000). Further polishing was conducted for 4 min on cloths with 1 μ and 0.3 μ Alpha alumina and 2 min with 0.02 μ colloidal silica.

2.3. Shell microstructure

The polished smooth surface was etched for 5 min using 5 mM ethylenediaminetetraacetic acid (EDTA) for SEM examination (Chan et al., 2012; Fitzer et al., 2014b). After etching, the blocks were mounted and coated with the revealed smooth surfaces mounted upwards on aluminum stubs using carbon tape and painted with silver gel to enhance the conductivity, and then sputter coated with carbon (~50 nm). The microstructures of the oyster shells were observed using on the LEO-1530 Field Emission Gun-Scanning Electron Microscope (FEG-SEM) with a beam voltage of 20 kV (Zeiss, Germany) under an accelerating voltage of 20 kV. The cross-sectional porosity of the foliated layers was calculated using ImageJ software by standardizing and converting an SEM image using thresholding (Hartig, 2013). The pore area was then calculated by using the ImageJ “Analyse Particles” feature due to the divergence in the size of pores. The pore area was measured with a confidence area of $> 0.001 \mu\text{m}^2$. The porosity thresholding was calculated using the non-diffracted regions of images produced by backscattered electrons. Three randomly selected specimens were examined per treatment (one individual per replicate). Kruskal-Wallis tests were used to determine the effect of pH on porosity.

2.4. Crystallographic orientation

Resin blocks without etching were mounted and coated followed the same methodology as described in the previous SEM section. The EBSD measurements were carried out on the LEO-1530 Field Emission Gun-Scanning Electron Microscope (FEG-SEM) with a beam voltage of 20 kV (Zeiss, Germany) with the stage tilted to 70°. For the EBSD measurement, a step size of 0.5 μm was used. The EBSD data were post-processed using the HKL-Channel 5™ software package and visualized by plotting the crystallographic orientation maps and pole figures that showed the extension of crystallographic orientation. Each color in the maps represents a corresponding crystallographic orientation according to the color key. Three randomly selected specimens were examined per treatment (one individual per replicate).

2.5. Computation of calculated stiffness

The calculated average stiffness of polyphase aggregates of calcitic crystal units in foliated layers with the direction perpendicular to the sectioned surface was computed for each specimen ($n = 3$) by MTEX, a freely available Matlab® toolbox for quantitative texture analysis (Mainprice et al., 2011) with the anisotropic elastic tensor of single-crystal calcite (Chen et al., 2001) and the crystallographic texture of calcite determined by EBSD based on the same coordinate system.

2.6. Shell mechanical properties

After obtaining a smooth surface from the specimens, load and displacement nanoindentation tests were conducted to determine the mechanical properties of the shell, i.e. hardness and the stiffness (elasticity) (Perez-Huerta et al., 2007). Briefly, nanoindentation tests were carried out using the Hysitron TriboIndenter TI 900 (TI 900, Hysitron, MN, USA) equipped with a Berkovich indenter (a half-angle of 63.5°). Hardness and stiffness values were calculated based on the

loading-unloading curve of each indentation using the Oliver-Pharr model (Doerner and Nix, 1986; Oliver and Pharr, 1992) (Fig. S1). The foliated layer in the middle region of the shell was selected as the region of interest to correspond the SEM and EBSD analysis. About 6–10 indents following a matrix-pattern were conducted with a loading of 9000 μN for each. Six specimens of each treatment were randomly selected for nanoindentation tests (two individuals per replicate). Measurements were firstly averaged within each specimen and then by replicate tank. Finally, three values per treatment were obtained ($n = 3$). All data were tested for normality of residuals, normality, and homogeneity of variance before analyzing by ANOVA. Student-Newman-Keuls test was used to compare the means following one-way ANOVA (Chan et al., 2012).

2.7. Shell mineral density and 3D surface density distribution

The surface topography and the mineral density according to three-dimensional (3D) digital information of the whole oyster juvenile shells were acquired using a high-resolution micro-CT scanning system (SkyScan 1076, Skyscan, Belgium) with a spatial resolution of 9 μm . Each individual air-dried oyster shell was placed on a plastic substrate and put in the chamber of micro-CT scanner. Around 300 two-dimensional (2D) layers were generated from each sample and used to examine the shell density and the volume ratio of partial density after standardization by the phantoms, which is used for bone density calculation, using the analytical software CT-Analyser v 1.14.4.1 (SkyScan, Kontich, Belgium) (Celenk and Celenk, 2012). Reconstruction software CT-Volume v 2.2.1.0 (SkyScan, Kontich, Belgium) was used to convert the 2D layers into a 3D reconstructed model. Three density ranges (0 to 1 g/cm³, 1 to 2 g/cm³ and higher than 2 g/cm³) were selected to represent the high, medium and low density ranges. The volume ratio of the three density ranges were calculated as Volume ratio = Volume of certain density range/Whole shell volume, to investigate the effect of the decreased pH on the shell construction with different density. Three randomly selected specimens (one individual per replicate) were used per treatment ($n = 3$). All data were tested for normality of residuals, normality, and homogeneity of variance before analyzing by ANOVA. Student-Newman-Keuls test was used to compare the means following one-way ANOVA.

3. Results

3.1. Shell microstructural damage in varying degrees under decreased pH

Irrespective of treatment level, decreased pH resulted in easily observable damage on the outer-most calcitic prismatic layer and the underlying aragonitic myostracum but had a less obvious impact on the “prominent” calcitic foliated layer (Fig. 1). The mineral erosion or dissolution was conspicuous on the outer prismatic layer on both the older hinge region and the younger middle region at pH 7.6 and pH 7.3 (Fig. 1A–F). Indeed, at the lowest pH, pH 7.3, the prismatic layer was absent from the older hinge and younger middle regions (Fig. 1C and F). Similarly, the impact of decreased pH on the myostracum layer which consisted of needle-like aragonitic prisms was also prominent (Fig. 1G–I). The myostracum prism was shorter and smaller in shape and appeared to be corroded with rounded edges in the treated juvenile shells at pH 7.6 and 7.3 compared with the intact needle-like prisms in the control (Fig. 1G–I). However, the impact of decreased pH on the foliated layer, the dominant structure occupied the majority of oyster shell and consisted of calcitic laths, was not clear (Fig. 1J–L). The foliated laths were well formed in thickness of approximately 200 nm with a laminated architecture in both decreased pH treatments and control individuals (Fig. 1J–L). At the lowest pH 7.3, sporadic gaps were observed between the foliated laths, while those of the shells under the control and the pH 7.6 were packed tightly. In fact, the area porosity of the foliated layer was not affected by the sporadic gaps ($\chi^2_{(2)} = 0.801$,

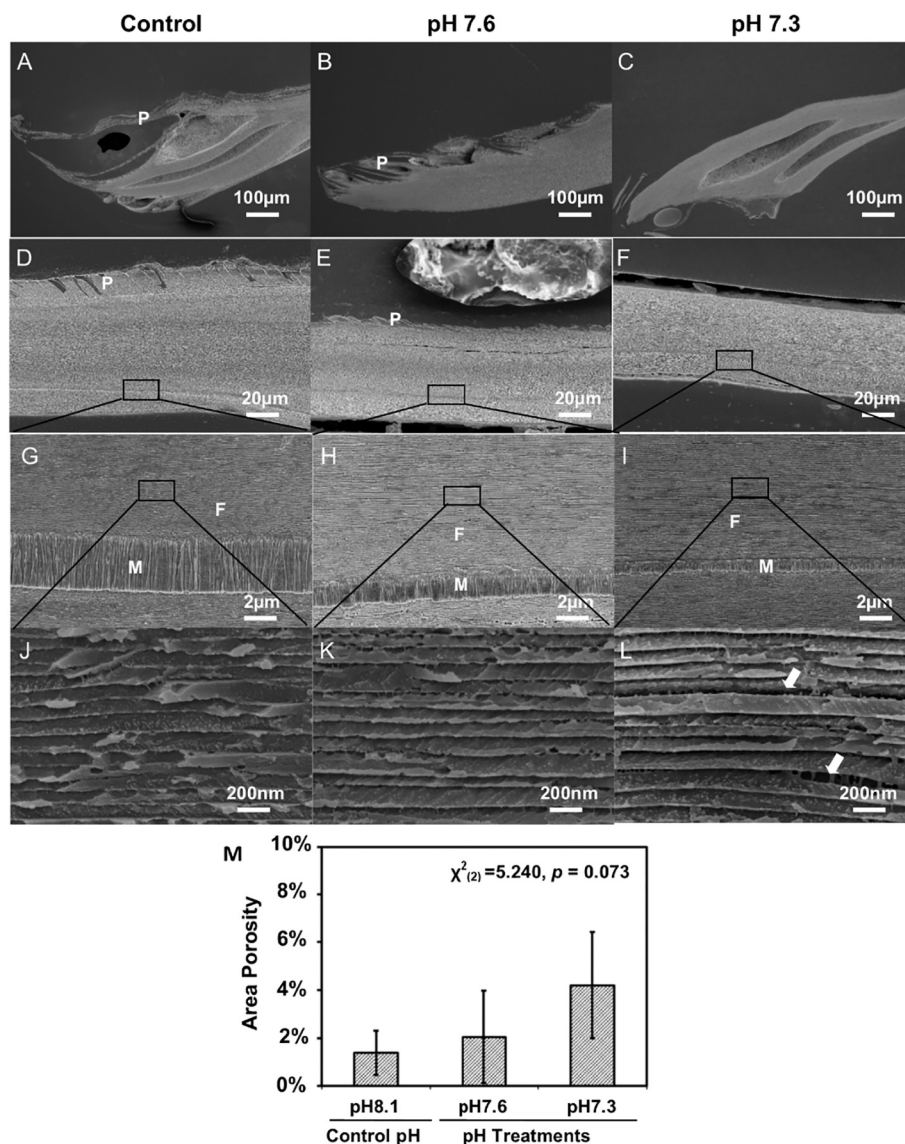


Fig. 1. Scanning electron micrographs from the cross-sectional surfaces of juvenile *Magallana hongkongensis* shells cultured in the control pH 8.1 (A, D, G and J), treatment pH 7.6 (B, E, H and K) and pH 7.3 (C, F, I and L) were compared. First row: scanning electron micrographs taken near the older hinge region (A–C). Second row: scanning electron micrographs taken near the younger middle region (D–F). Compare with the prismatic layer in the control (A and D), the prismatic layers have partially lost at pH 7.6 (B and E) and were not detectable at pH 7.3 (C and F) on both regions. Third row: well-formed myostracum at the control pH of 8.1 (G) and narrower myostracum with stunted aragonite prisms at pH 7.6 (H) and pH 7.3 (I). Fourth row: enlarged views of the foliated layer at the control (J), pH 7.6 (K) and pH 7.3 (L). (M): The area porosity of foliated layers under the control and decreased pH conditions. The mean values are presented in the bar chart (mean $\pm$ SD, $n = 3$). Annotations: P-prismatic layer; M-myostracum; F-foliated layer. White arrow: irregular gaping of the foliated layer.

$p = 0.073$) (Fig. 1M). Meanwhile, the chalky layer was not observable in the longitudinal section-surfaces of the juvenile oyster shells from both the control and the decreased pH treatments.

3.2. Less aligned calcite crystal units in foliated layers under decreased pH

The impact of the decreased pH on the crystallographic orientation of the foliated layer is represented by the color change in EBSD crystallographic orientation map (Fig. 2i and D). The spread of the data points in the pole figures indicate variation of the c -axis of the calcite crystal units (Fig. 2ii). In general, the crystallographic c -axis across the foliated structure is slightly tilted in the growth direction (elongation of the foliated lath) in both decreased pH treated and control juvenile shells. However, compared with the data points that aggregated with identical orientation from the control shells, there is clearly less constraint on the crystallographic c -axis of the calcite units in the juvenile shells grown under both decreased pH treatments (Fig. 2ii).

3.3. Decreased pH reduced mechanical properties

Two mechanical properties of shells, hardness and stiffness, were quantitatively measured on the foliated layer using nanoindentation, hereafter as “experimental” hardness and stiffness in Fig. 3. Notably,

decreased pH treatment levels used in this study did not affect hardness, when compared to control ($F_{(2,6)} = 3.931, p = 0.081$) (Fig. 3A). Although decreased pH 7.3 significantly reduced the shell stiffness when compared to control shells, the shell stiffness under pH 7.6 was unchanged ($F_{(2,6)} = 6.903, p = 0.028$) (Fig. 3B). Importantly, the effective stiffness calculated based on the crystallography obtained by EBSD, hereafter as “calculated” stiffness in Fig. 3, showed a significant decrease of value at pH 7.3 compared with that at the control pH ($F_{(2,6)} = 5.945, p = 0.038$), which agreed with that of the experimental stiffness (Fig. 3B).

3.4. Mineral density distribution under the decreased pH

Unlike previous measurements which focus on specific points or regions, the whole shell was assessed for the impact of decreased pH on hard structural features using micro-computed tomography (Micro CT) (Fig. 4). The 3D reconstructed model showed that the newly formed edge region and the surface were loaded with low-density materials as represented in red (Fig. 4i). Notably, the interior surfaces of shells formed at pH 7.3 and pH 7.6 were loaded with medium- and low-density materials as shown in pink and red colors (Fig. 4ii).

The shells fell into three density categories, i.e. $< 1 \text{ g/cm}^3$, $1\text{--}2 \text{ g/cm}^3$ and $> 2 \text{ g/cm}^3$. The volume ratio of density $< 1 \text{ g/cm}^3$ in the two

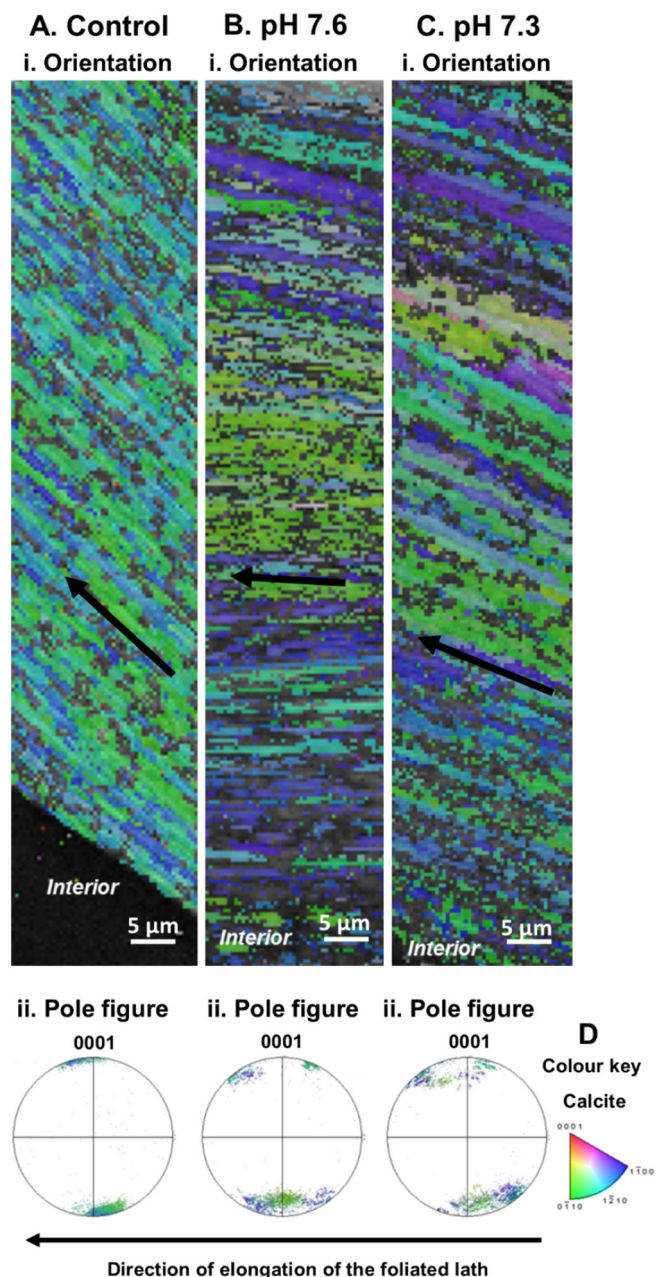


Fig. 2. Electron backscatter diffraction analyses of shells grown at ambient or control pH 8.1 (A), treatment pH 7.6 (B) and pH 7.3 (C). Crystallographic orientation map (i) of calcite crystals of shell foliated layer in reference to the {0001} plane. Crystallographic planes of calcite are color-coded linked to the normal crystallographic direction using the color key (Perez-Huerta and Cusack, 2009) (D). Pole figures for calcite (ii) with the direction of elongation of the foliated lath correspond to the crystallographic orientation maps and used the same color key. Black arrow: the direction of elongation of the foliated layer. Scale bar = 5 μm.

decreased pH treatments (pH 7.6 and pH 7.3) was significantly increased compared when compared to control ($F_{(2,6)} = 7.120$, $p = 0.026$) (Fig. 4D). There was no significant difference in the volume proportions of the middle (1–2 g/cm³) and high (> 2 g/cm³) shell mineral density categories as well as the shell mineral density among pH treatments (Fig. 4D and E).

4. Discussion

This study clearly demonstrated the modulating effect of

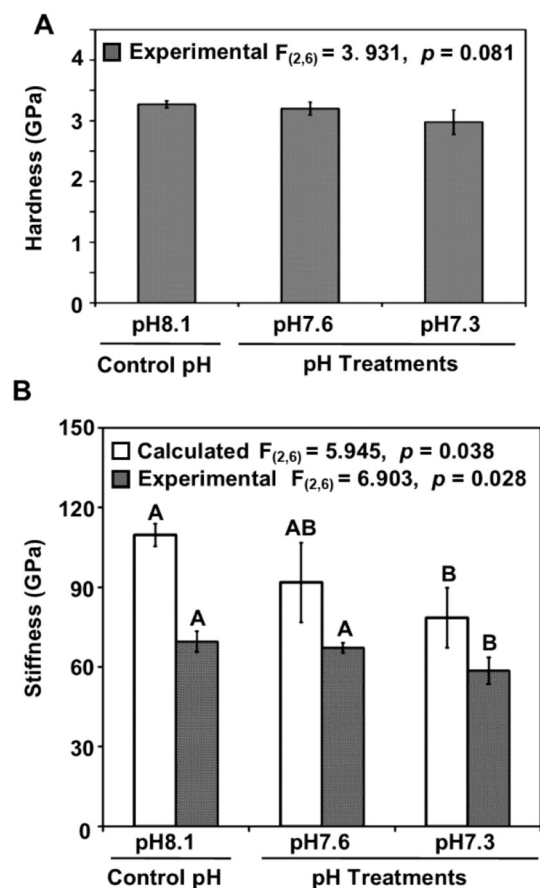


Fig. 3. The impact of the decreased pH conditions on the shell mechanical properties in terms of hardness (A) and stiffness (B) of *Magallana hongkongensis* from the cross-sectional surfaces were compared. The experimental hardness and stiffness were measured by nanoindentation in cross-sectional shell surfaces of *Magallana hongkongensis*. The calculated stiffness was computed based on the crystallography in the same direction. Data of mechanical properties are presented as mean $\pm$ SD of three replicates ($n = 3$). Different capital case letters illustrate significant differences ($p < 0.05$) among treatments found through comparisons Student-Newman-Keuls test following a one-way ANOVA.

environmental pH and carbonate chemistry associated with OA on structural integrity, crystal unit orientation and mechanical characteristics of an oyster shell. Even a moderate decrease in pH caused structural impairment on the protective outer and inner shell layers, i.e., prismatic and myostracum layers, respectively. In contrast, the inner foliated layer, which makes up the bulk of the shell, was microstructurally intact with “irregularly” aligned calcite crystal units at pH 7.6 and 7.3. The altered crystallography ultimately leads a decrease in stiffness relative to the control, but only in the undersaturated treatment (pH 7.3, $\Omega_{\text{Cal}} \sim 0.78$), which coincided with the results of experimental stiffness obtained by nanoindentation. These results provide insight into the biomineralization adaptations of estuarine oysters to tolerate unfavorable calcification conditions, which will be discussed below by revealing the possible linkages between the observed microstructural and mechanical features and also comparing the results with other related studies.

4.1. Structural impairment of oyster shells under decreased pH stress

Except for the invisible chalky layer in the juvenile oyster shells, the other three shell layers, the prismatic, foliated and myostracum exhibited impairment of shell microstructures to varying degrees under decreased pH conditions. Firstly, as expected, the prismatic layer started to dissolve at pH 7.6 and it was completely absent at pH 7.3

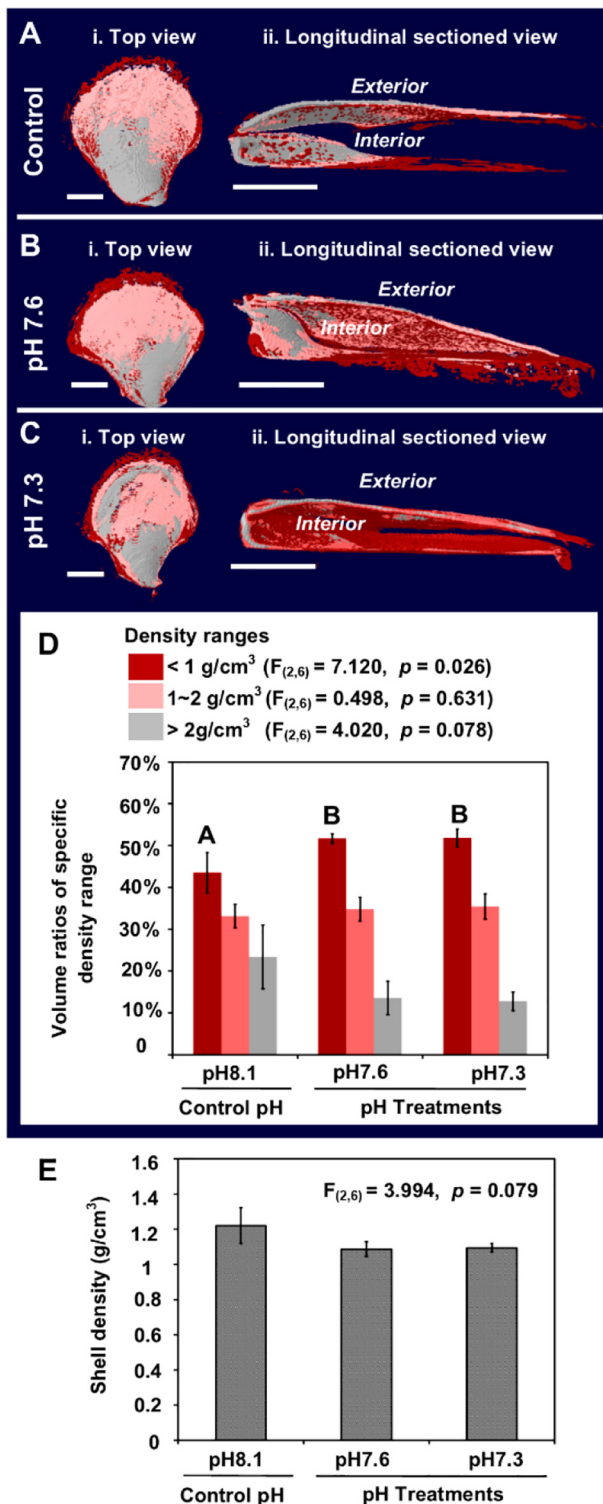


Fig. 4. The impact of the decreased pH conditions on the shell density of *Magallana hongkongensis* cultured under at ambient or control pH 8.1 (A), treatment pH 7.6 (B) and pH 7.3 (C) from the top view (i) and the longitudinally sectioned view (ii) are reconstructed and showed representative by the micro-CT scans. The impact of the decreased pH conditions on the shell volume ratio of partial density categories (D), and overall density (E) are examined. The volume ratio of partial density categories was quantified, i.e. < 1 g/cm^3 , 1–2 g/cm^3 and > 2 g/cm^3 . The mean values were presented in a bar chart (mean $\pm$ SD, $n = 3$). Scale bar = 3 mm.

(Fig. 1A–F). Similar impacts of decreased pH on the prismatic layer were reported for the juvenile scallop, *Argopecten irradians* (Talmage and Gobler, 2010), juvenile hard-shell clams, *Mercenaria mercenaria* (Dickinson et al., 2013) and the rock oyster, *Saccostrea glomerata* (Watson et al., 2009). For outermost mineralized layer, the prismatic layer, dissolution occurred at pH 7.6 with a supersaturated state of calcite ($\Omega_{\text{Cal}} \sim 1.6$). Similarly, the dissolution of the prismatic layer in a pteropod shell not only occurred first but also in similarly supersaturated environments ($\Omega_{\text{Ar}} \sim 1.65$) (Bednaršek et al., 2012). Therefore, the dissolution of the prismatic layer in this study may be regarded as an initial observable bioindicator of oysters experiencing decreased pH environments.

Secondly, in the myostracum layer the aragonite prisms which are the building blocks were significantly stunted at pH 7.6 ($\Omega_{\text{Ar}} \sim 0.96$) and pH 7.3 ($\Omega_{\text{Ar}} \sim 0.47$) with an undersaturated state of aragonite. Although the foliated layers were formed with normal microstructures, sporadic gaps between the lath were observed in the calcite undersaturated treatment (pH 7.3, $\Omega_{\text{Cal}} \sim 0.78$). According to the molluscan shell calcification process, the mineralization process occurs in compartments with supersaturated conditions at the shell edge (Mount et al., 2004). This means that the newly formed myostracum (or foliated layer), located in the innermost shell surface, is directly exposed to the environment, which may be corrosive and undersaturated (Mount et al., 2004; Toyofuku et al., 2017). The exposed aragonite prisms (or calcite foliated lath) may have been subjected to partial dissolution of the formed crystal units. Similar results were also reported and characterized for the pteropod shell (Bednaršek et al., 2012), mussels (Hahn et al., 2012; Fitzner et al., 2014a) and tubeworms (Li et al., 2014). Therefore, the underlying biomineralized structures with partly eroded building blocks can be regarded as another bioindicator for oysters, representing a more severely deteriorated state caused under OA scenarios.

In addition, these two indicators were verified at a scale that encompassed the entire shell by micro-CT scanning. The overall shell density was not affected, which corroborates the unchanged area porosity of the well-formed foliated layer. However, the increased low-density volume under decreased pH indicates that shell dissolution may happen for the entire shell, especially for the inner surface that had a density < 1 g/cm^3 in pH 7.3, while it was mostly > 2 g/cm^3 in the control (Fig. 4). During the mineral formation process, marine invertebrates are capable of actively increasing the site of calcification by pumping proton out of the calcification site, thereby enabling calcium carbonate precipitation (Toyofuku et al., 2017). Supersaturated calcite conditions in oyster were found in the restricted compartment around the shell edge including the outer mantle and the first intracellular nucleation site (Mount et al., 2004). Undersaturated calcite conditions may be maintained elsewhere in contact with the inner shell surface (Addadi et al., 2006; Thomsen et al., 2010). This process was recently confirmed in foraminifera, where a dramatic decrease of pH appeared near the nucleation site (Toyofuku et al., 2017).

Therefore, in low pH conditions due to OA, these inner areas of newly formed minerals, which are precipitated as structural building blocks for the prismatic and foliated layers, may still be prone to dissolution. When the shell dissolution rate is faster than the mineralization rate, organisms tend to produce thinner and lighter (less dense) shells resulting in impaired shell microstructure. This may explain the multiple negative effects of reduced pH in our results, including porous and less dense foliated layers.

In this way, the inner surface of the oyster shell may be more prone to dissolution because of the decreased pH environment in the calcification compartment or mantle fluid underneath those layers (Melzner et al., 2011). Similar to this study, the serpulid tubeworm, *Hydroides elegans*, showed impaired tube structure with significantly lower density minerals when exposed to decreased pH 7.8 (Li et al., 2014; Li et al., 2016).

4.2. The shell stiffness is highly related to the crystallography under decreased pH

Although there were relatively dramatic damages in the prismatic and myostracum layers under OA scenarios, the foliated layer was formed with well-maintained laminated architecture and complete foliated lath irrespective of treatment levels. However, the calcite crystal units were significantly less aligned in the foliated layer at pH 7.6 and pH 7.3. Marine organisms build their skeletons and shells through aligning crystal units into well-evolved architectures, and the microstructure, organization and orientation of these crystal units determine the ultimate mechanical properties, i.e. stiffness of shells (Rodríguez-Navarro et al., 2002; Agbaje et al., 2017). The foliated layer is the dominant structure that takes up the majority of this estuarine oysters shell, it consist of calcite crystal units in elongated lath which are separated by small amounts of organic membrane, which is critical for mechanical support and protection (Lee et al., 2011). In order to test if the sporadic gaps affected the entire structure under decreased pH conditions, we quantified the area porosity of the foliated layers. It was showed the area porosity was not affected at low pH, indicating the defects (gaps) of the foliated layer were insufficient to alter the compactness (Fig. 1M) of the microstructure and thus the corresponding effectiveness of microstructure on mechanical properties. It may be due to the calcite-based components and their interior position which made the foliated layers less susceptible to the decreased pH conditions, separated from contact with the seawater by the protective prismatic layer. However, the well-formed architecture in the foliated layer is marred by less constrained calcite crystal units, in terms of their crystallographic orientation, at pH 7.6 and 7.3 (Fig. 2). Although the impact of OA on crystallography is not well studied, these results are consistent with those of the study in which the blue mussel, *Mytilus edulis* grown under pH 7.5 (750 μ atm) and pH 7.3 (1000 μ atm) showed less constrained crystallographic orientation compared to those grown in ambient, pH 7.7 (380 μ atm), conditions (Fitzer et al., 2014b). Given the high anisotropic stiffness (elastic modulus) of an individual calcite crystal (Chen et al., 2001) and the determining role of crystallography on mechanical properties, the impact of the observed decrease in alignment on shell stiffness was revealed based on the altered crystallography observed in this study. In the same direction perpendicular to the sectioned surface, the changes in the calculated stiffness coincided with the experimental stiffness measured by nanoindentation (Fig. 3B), indicating that shell stiffness depends on the aggregation of underlying crystal units. Therefore, based on the above analyses, the impact of decreased environmental pH on the stiffness of foliated layer can be explained by the retained area porosity and the stiffness calculated using crystallography data.

The organic matrix is also a key factor, in addition to microstructure and crystallography, in determining the mechanical properties of the shell (Marin and Luquet, 2004). The involvement of organic constituents in low proportions (0.01 to 5 wt.%) increased the shell fracture toughness of the conch, *Strombus gigas*, by 2 to 3 orders of magnitudes (Kamat et al., 2000). However, the contribution of the organic matrix to the stiffness of the foliated layer in oysters is limited at the micro-meter (nanoindentation) scale and even less than its contribution to hardness. After removing all the organic matrix from the foliated layer of Pacific oyster, *Magallana gigas* (previously *Crassostrea gigas*), the hardness decreased by approximately 27% and the stiffness (elastic modulus) decreased only by 20% (Lee et al., 2008c). Moreover, the density of organic matrix is highly varied across the different shell components within the oyster shell (Dauphin et al., 2013). It is necessary to separate the different shell components to evaluate the contribution of organic matrix to the corresponding mechanical properties of the shell. However, because of the irregular arrangement of the calcitic shells of *Magallana* and the small size of the juvenile shells used in this study, it is not possible to separate the different layers to quantify their mechanical properties and organic components. A future longer-term

experiment is needed to harvest shells large enough to investigate the specific matrix proteins packing in each shell structure at the micro or nano-spatial scale, and to illustrate the relationship between shell proteins and the mechanical properties under OA.

4.3. A possible compensatory mechanism of *M. hongkongensis* and the ecological implications

The result present here showed the shell is damaged to varying degrees on the different structural components of oyster shells, some as sensitive bioindicators showing exposure decreased pH conditions and indicating the inevitable deterioration under OA scenarios. However, the microstructure of the major shell component, the foliated layer, was well-maintained in terms of the fundamental laminated architecture and the integrity of the individual foliated laths used as building blocks. Interestingly, the formation of normal foliated structures was accompanied by a decrease of biological crystallographic control. In foliated layer formation, a well-defined foliated lath is formed by unoriented nanocrystals that contain granulocytic haemocytes (Mount et al., 2004; Johnstone et al., 2015), this is an energy demanding process. With lowered metabolic rates caused by the decreased pH conditions (Dupont and Portner, 2013), energy shortages may occur in oysters, which may limit their control of precipitation, and as we showed, result in the less aligned and controlled calcite crystal units in the foliated layers, which was also apparent in the blue mussel (Fitzer et al., 2014a). Meanwhile, although the altered crystallography ultimately lead to a decrease of stiffness under the most acidified environment at pH 7.3 ($\Omega_{\text{cal}} \sim 0.78$), the hardness and stiffness were both retained at pH 7.6 even though the calcite crystal units were also relatively less aligned. Therefore, these less aligned or controlled crystal units may be a compensatory mechanism of *M. hongkongensis*, an estuarine oyster species, to maintain the fundamental microstructure and mechanical properties of the folia layer while at the same time conserving energy under OA scenarios. This compensatory mechanism may be an adaptation of coastal organisms that experience high fluctuation in coastal and estuarine environments. Similarly, the larvae of the rock oyster, *S. glomerata*, from adults exposed to increasing $p\text{CO}_2$ associated with OA can withstand larger changes than the larvae of wild adults (Parker et al., 2011). Similarly, the results from several other recent studies suggest that organisms living within upwelling environments, such as coastal and estuarine environments, show a degree of adaptation to OA (Wright et al., 2014; Lardies et al., 2017). Consequently, with such adaptations to compensate for fluctuating environments, *M. hongkongensis* in its current state may be sustainable as the dominant species along the southern coast of China.

Nevertheless, given the continuing seawater acidification scenarios and the high fluctuation of seawater chemistry in estuaries (including pH), the survival of *M. hongkongensis* is uncertain in future coastal oceans as the extreme decrease of pH continues. At the lowest pH of 7.3, the less aligned calcite crystals ultimately induced a significant decrease of stiffness, which indicated that the effectiveness of the compensatory mechanism of *M. hongkongensis* can be diminished in more-acidic seawater resulting from continuing OA scenarios. A calcite-based shell is brittle in nature, like egg shells or ceramics, therefore their resistance to deformation (or breaking force) largely depends on its stiffness parameter. The less stiff foliated layer at pH 7.3 may imply an increased susceptibility of these oysters to predators under OA scenarios (Sanford et al., 2014). The initiation of resistance to predation occurs at the post-settlement larval stages and their defences are vital to the oyster population (Newell et al., 2007). The juvenile Olympia oyster, *Ostrea lurida* formed a weaker shells under pH 7.8 and were more prone to being crushed by their predator (Sanford et al., 2014). Consequently, under a predictable long-term scenario, OA may pose a non-negligible threat to the sustainability of the coastal oyster industry and the health of the oyster reef structure.

Conflict of interest

The authors declare no competing or financial interest.

Author contributions

YM and VT conceived and designed the study. ZG and HY performed the nanoindentation test and contributed with materials and analysis equipment. YM performed the CO₂ propagation experiment, SEM and EBSD analyses and contributed to the analysis equipment. KY contributed with materials and analysis equipment for Micro-CT scanning. YM analyzed the data. YM and VT drafted the manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2018.12.030>.

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